Exp No: 18

GENERATION OF PWM USING TIMER INTERRUPT

AIM:
To generate PWM using Timer interrupt.

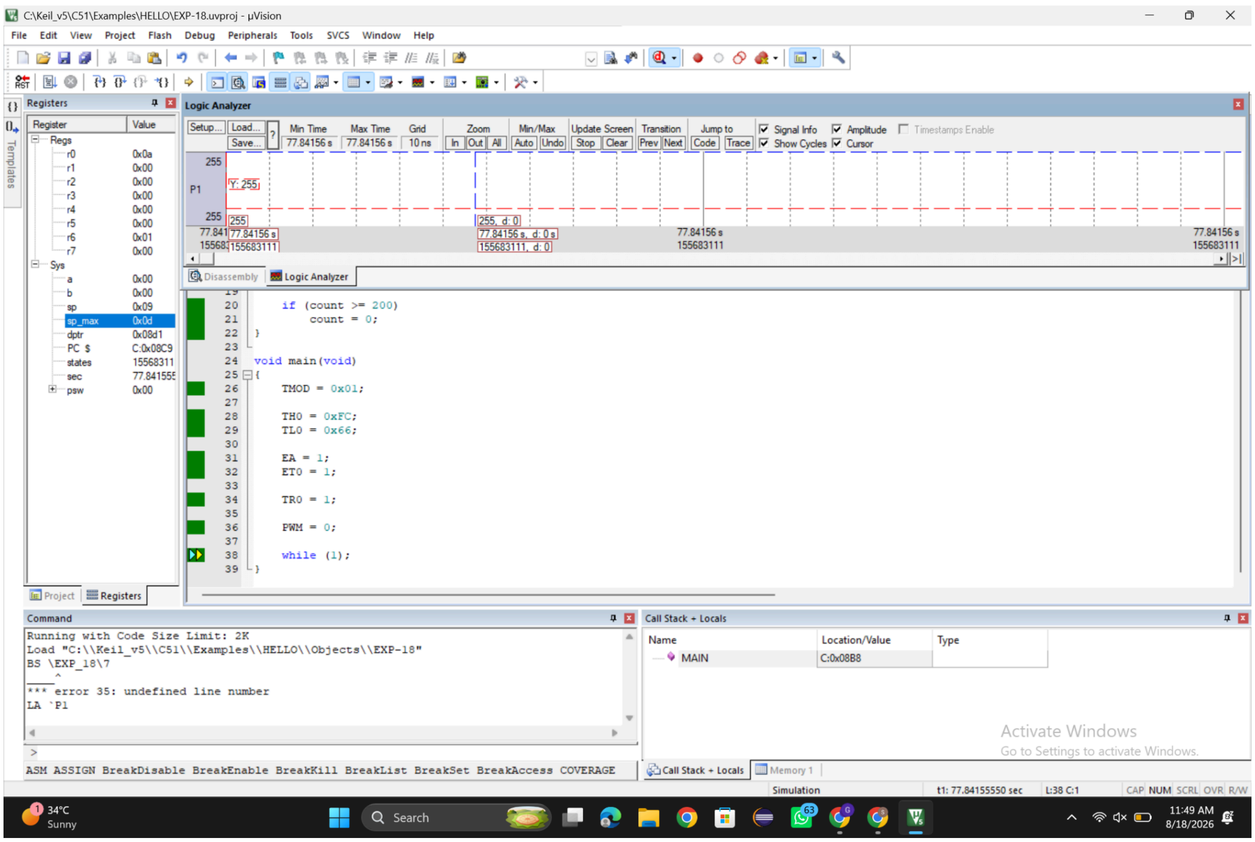
APPARATUS / SOFTWARE REQUIRED:
• PC with Windows OS
• Keil µVision IDE
• 8051 Simulator (Keil built-in)
• Serial Window (UART #1)

PROGRAM:
```
#include <reg51.h>

sbit PWM = P1^0;

unsigned char count = 0;
unsigned char duty = 100; // Change this value (0–200)

void timer0_ISR(void) interrupt 1
{
TH0 = 0xFC; // Reload for 1ms delay
TL0 = 0x66;

count++;

if(count < duty)
PWM = 1; // ON time
else
PWM = 0; // OFF time

if(count >= 200) // Total period = 200ms

count = 0;
}

void main()
{
TMOD = 0x01; // Timer0 Mode 1 (16-bit)
TH0 = 0xFC; // 1ms preload
TL0 = 0x66;
```

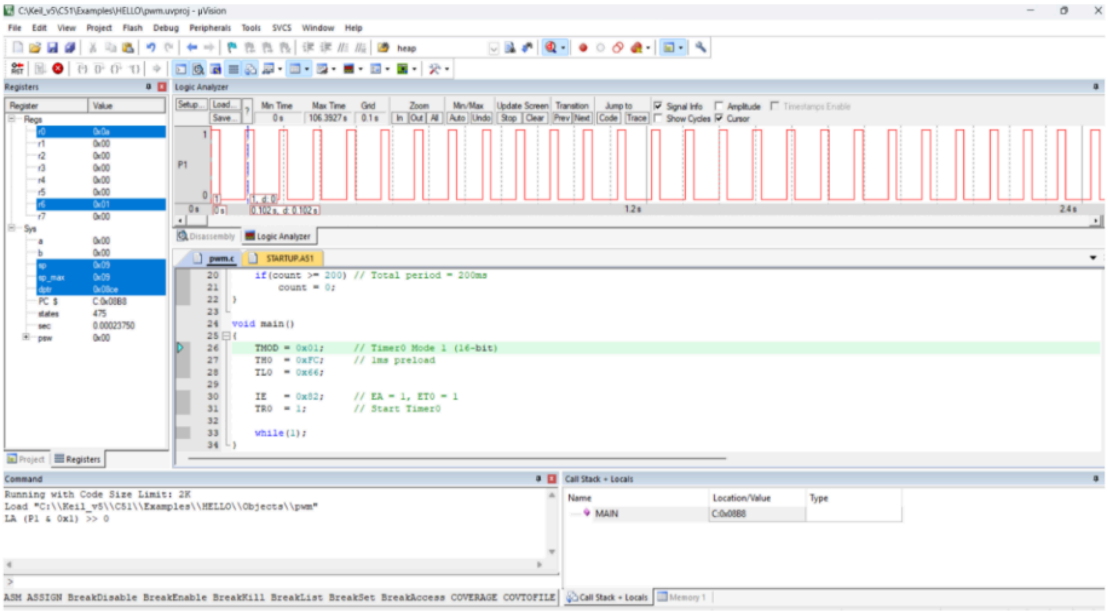
```
IE = 0x82; // EA = 1, ET0 = 1
TR0 = 1; // Start Timer0

while(1);
}
```

OUTPUT:

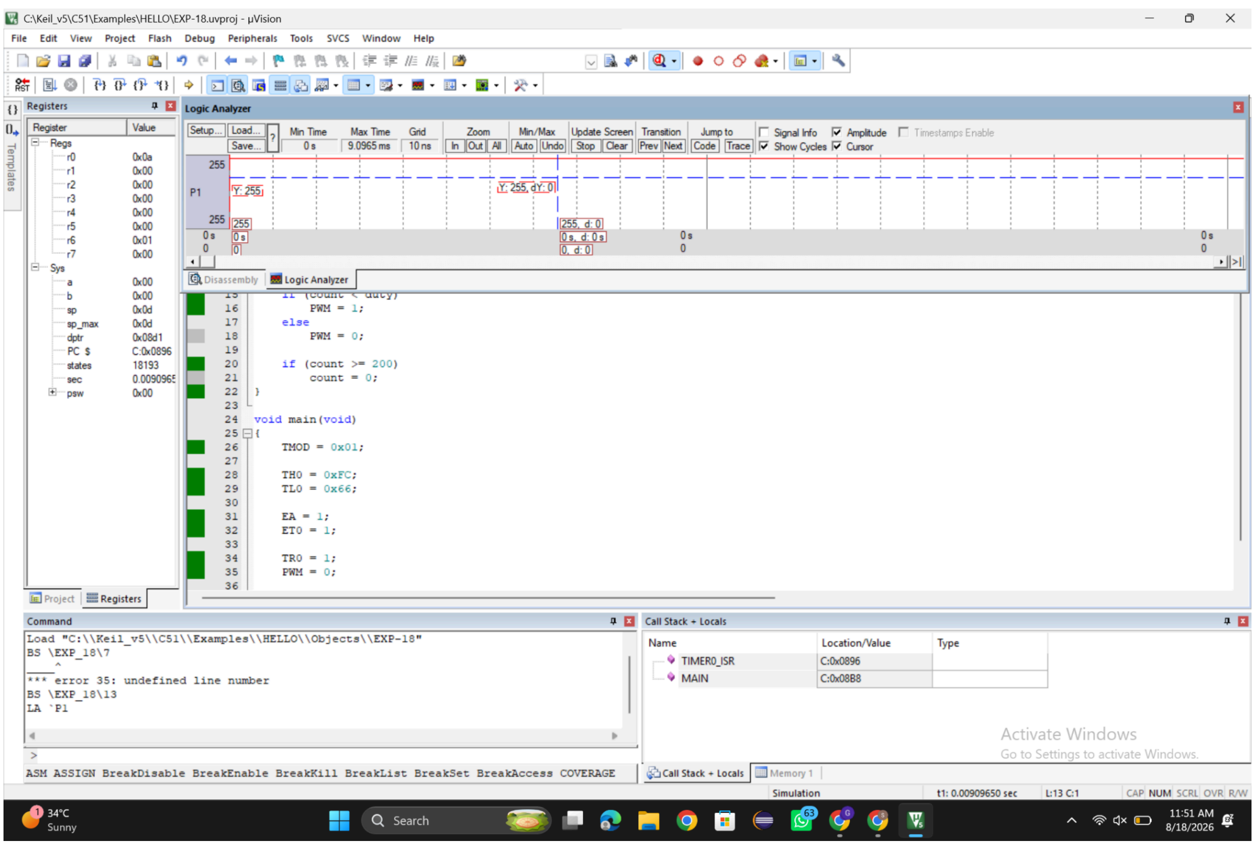
50% Duty cycle:



25% duty cycle:



75% duty cycle:



Formula:

Duty cycle (%) = (ON time / Total period) × 100

RESULT:
Thus the PWM was generated using Timer Interrupt for various duty cycle.